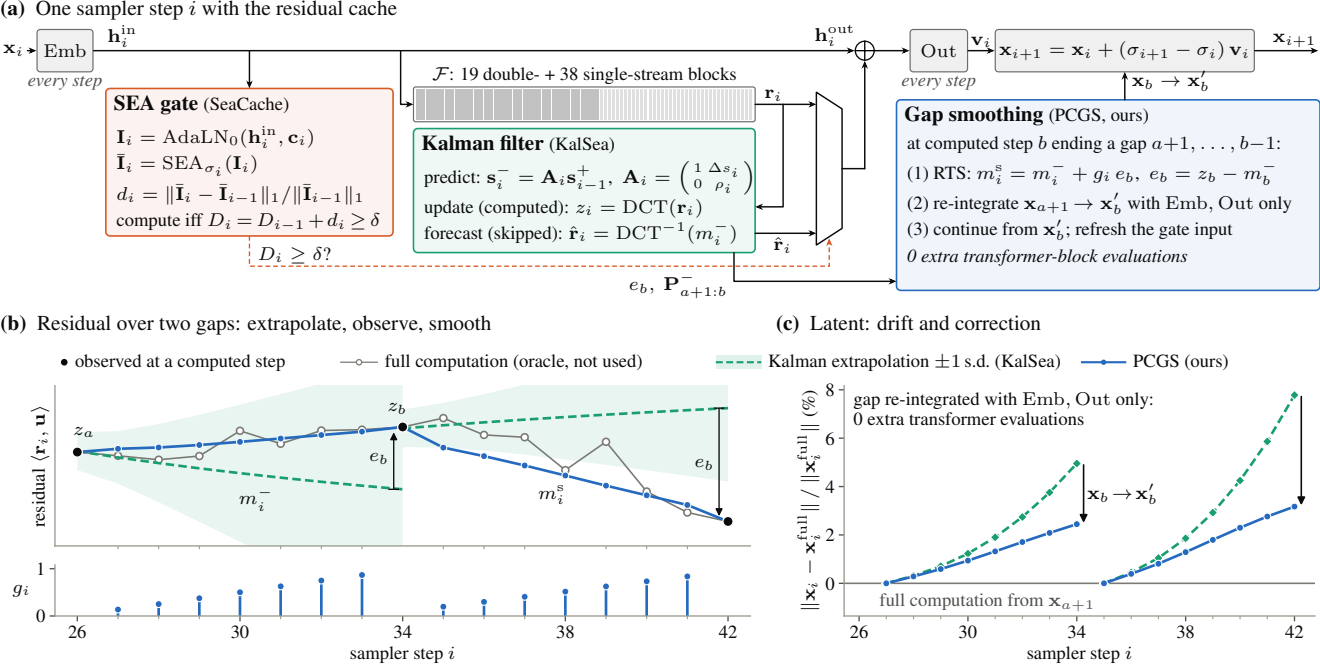

(a) One sampler step i with the residual cache
$\mathbf{x}_i$
Emb
$\mathbf{h}_i^{\text{in}}$
every step
SEA gate (SeaCache)
$\mathbf{I}_i = \text{AdaLN}_0(\mathbf{h}_i^{\text{in}}, \mathbf{c}_i)$
$\bar{\mathbf{I}}_i = \text{SEA}_{\sigma_i}(\mathbf{I}_i)$
$d_i = \|\bar{\mathbf{I}}_i - \bar{\mathbf{I}}_{i-1}\|_1 / \|\bar{\mathbf{I}}_{i-1}\|_1$
compute iff $D_i = D_{i-1} + d_i \geq \delta$
$D_i \geq \delta$?
$\mathcal{F}$: 19 double- + 38 single-stream blocks
$\mathbf{r}_i$
Kalman filter (KalSea)
predict: $\mathbf{s}_i^- = \mathbf{A}_i \mathbf{s}_{i-1}^+$, $\mathbf{A}_i = \begin{pmatrix} 1 & \Delta s_i \\ 0 & \rho_i \end{pmatrix}$
update (computed): $z_i = \text{DCT}(\mathbf{r}_i)$
forecast (skipped): $\hat{\mathbf{r}}_i = \text{DCT}^{-1}(m_i^-)$
$\hat{\mathbf{r}}_i$
$\mathbf{h}_i^{\text{out}}$
Out
$\mathbf{v}_i$
every step
$\mathbf{x}_{i+1} = \mathbf{x}_i + (\sigma_{i+1} - \sigma_i)\,\mathbf{v}_i$
$\mathbf{x}_{i+1}$
$\mathbf{x}_b \to \mathbf{x}'_b$
Gap smoothing (PCGS, ours)
at computed step b ending a gap $a+1, \ldots, b-1$:
(1) RTS: $m_i^{\text{s}} = m_i^- + g_i\, e_b$, $e_b = z_b - m_b^-$
(2) re-integrate $\mathbf{x}_{a+1} \to \mathbf{x}'_b$ with Emb, Out only
(3) continue from $\mathbf{x}'_b$; refresh the gate input
0 extra transformer-block evaluations
e_b, $\mathbf{P}_{a+1:b}^-$
(b) Residual over two gaps: extrapolate, observe, smooth
observed at a computed step
full computation (oracle, not used)
residual $(\mathbf{r}_i, \mathbf{u})$
z_a
z_b
e_b
m_i^-
m_i^{s}
g_i
sampler step i
(c) Latent: drift and correction
Kalman extrapolation ± 1 s.d. (KalSea)
PCGS (ours)
gap re-integrated with Emb, Out only:
0 extra transformer evaluations
$\|\mathbf{x}_i - \mathbf{x}_i^{\text{full}}\| / \|\mathbf{x}_i^{\text{full}}\|$ (%)
$\mathbf{x}_b \to \mathbf{x}'_b$
full computation from $\mathbf{x}_{a+1}$
sampler step i